1. Introduction

Enterprise Resource Planning (ERP) systems have become an integral component of modern organizations by enabling the seamless integration of business processes, data, and information flow across multiple departments. In today's competitive and fast-paced business environment, organizations can no longer operate efficiently using isolated or standalone systems. ERP systems address this challenge by providing a unified platform that connects finance, operations, administration, and management functions into a single, coherent framework.

In large enterprises—particularly manufacturing and export-oriented organizations—the efficiency of operations is heavily dependent on the accuracy, consistency, and availability of foundational data maintained within the system. Business activities such as financial accounting, asset management, reporting, and compliance rely on predefined structures and reference data. Any inconsistency at this foundational level can lead to cascading errors across the organization, affecting productivity, financial accuracy, and strategic decision-making.

At the core of every ERP system lies master data, which defines the structural blueprint of enterprise operations. Master data includes critical information such as organizational units, financial groupings, account codes, asset classifications, banking details, and user credentials. This data is relatively stable but is referenced continuously by transactional processes and reporting mechanisms. As a result, master data acts as the backbone of all ERP functionalities, ensuring uniform interpretation of information across departments.

Poorly managed master data can result in serious organizational issues such as incorrect financial statements, duplicate records, operational delays, audit non-compliance, and unreliable management reports. Therefore, effective master data management is not merely a technical requirement but a strategic necessity for enterprise governance and operational control.

The ERP Master Module developed in this project aims to address these challenges by providing a structured, centralized, and scalable solution for managing critical master data. The system is designed using modern software technologies. By establishing a robust master data foundation, the project demonstrates how a well-designed ERP master module can significantly enhance data integrity, operational efficiency, and overall system effectiveness.

1.1 Importance Of The Project

The importance of this project lies in addressing one of the most critical challenges in enterprise systems—effective master data management. In ERP environments, master data serves as a single source of truth for all business operations. Accurate master data ensures consistency across departments and enhances the reliability of financial and operational reports.

This project is significant for the following reasons:

- Ensures centralized and standardized management of ERP master data
- Reduces data duplication and minimizes manual errors
- Improves operational efficiency through structured workflows
- Supports financial compliance and audit requirements
- Provides a scalable foundation for future ERP modules
- Enhances user productivity through a professional and intuitive interface

The project provides hands-on exposure to enterprise-level system analysis, architectural design, database modeling, and full-stack application development.

1.2 General Organization of the Report

The report is structured to progress logically from conceptual understanding to implementation details, providing a comprehensive guide for readers:

- Chapter 1 – Introduction: Presents the background, project importance, objectives, and organizational context.
- Chapter 2 – System Analysis: Discusses the limitations of existing systems, problem statements, proposed solutions, and system requirements.
- Chapter 3 – System Design: Explains architecture, database schema, workflows, and graphical user interface design principles.
- Chapter 4 – Project Description: Provides module-wise explanation, functional scope, and real-world operational scenarios.
- Chapter 5 – System Development: Details the technologies, tools, code logic, pseudocode, and development methodology..

1.3 Organization Profile

Ambattur Clothing Company (ACC) is a leading garment manufacturing and exporting organization that has established a strong presence in the global apparel industry since its inception in 1981. Beginning its operations with a single manufacturing unit, the company has grown consistently over the decades into a large-scale enterprise known for producing high-quality garments for several internationally renowned fashion brands. ACC's growth journey reflects its strategic vision, operational excellence, and unwavering commitment to quality and customer satisfaction.

The organization operates multiple state-of-the-art manufacturing facilities across India, each equipped with advanced machinery and modern production technologies. These facilities adhere to standardized manufacturing processes and stringent quality control measures to ensure uniformity, efficiency, and precision at every stage of production. ACC's robust operational framework enables it to meet international quality standards and maintain long-term partnerships with global clients.

ACC's strong global presence is further reinforced by an extensive supplier and logistics network that supports efficient sourcing of raw materials, streamlined production planning, and timely delivery of finished products. The company places significant emphasis on innovation, continuous process improvement, and the adoption of best manufacturing practices to remain competitive in the international market.

Beyond operational excellence, Ambattur Clothing Company is committed to sustainability, ethical manufacturing, and corporate social responsibility. The organization actively implements environmentally responsible practices, promotes employee welfare, and complies with international labor and safety standards. These initiatives not only contribute to sustainable growth but also strengthen the company's reputation as a socially responsible enterprise.

As a large-scale manufacturing organization, ACC manages complex business operations involving financial structures, asset management, organizational hierarchies, and banking relationships. Efficient management of such complexity requires reliable, accurate, and integrated information systems. Enterprise Resource Planning (ERP) systems play a critical role in supporting these requirements by enabling centralized data management, improved transparency, and effective decision-making.

2. System Analysis

2.1 Problem Definition

In today's rapidly growing and complex business environment, the management of ERP master data remains a critical challenge for many organizations. Despite its importance, numerous enterprises still rely on outdated, semi-manual, or fragmented systems that lack adequate standardization, validation, and centralized control. These legacy systems were initially designed to handle limited volumes of data and basic operational needs, which may have been sufficient for smaller organizations or simpler processes. However, as organizations expand in size, operational complexity, and geographic reach, these systems become increasingly inefficient, error-prone, and difficult to maintain. The growing volume of master data, coupled with inconsistent or unstructured management practices, frequently leads to duplicate records, conflicting classifications, missing information, and increased maintenance efforts.

Master data is the foundation of enterprise operations, defining organizational hierarchies, financial groupings, asset categories, supplier and customer profiles, and other critical reference points. Inaccurate, inconsistent, or outdated master data can directly disrupt essential business processes such as accounting, procurement, inventory management, asset tracking, reporting, and regulatory compliance. For example, duplicate account codes may result in incorrect financial statements, while inconsistent asset classifications can lead to errors in depreciation calculations and inventory reporting. Operational delays in production planning, procurement approvals, and supply chain execution may also occur, reducing overall efficiency.

Another key limitation of existing systems is the lack of scalability and flexibility. Modifying, updating, or restructuring master data often requires extensive manual intervention, which not only increases the risk of errors but also consumes significant time and resources. Weak security measures, limited role-based access, and the absence of comprehensive audit trails further compound governance and compliance challenges. These issues underscore the urgent need for a modern, centralized, and structured ERP master management system that ensures accurate, secure, and reliable data, while enhancing operational efficiency, supporting informed decision-making, and maintaining regulatory compliance across the organization.

2.2 Existing System

Limited automation, rigid structures, and a dependence on individual expertise largely characterize the current systems used for ERP master data management. Master data entry and maintenance are heavily manual, with minimal system-driven validation or standardization. As a result, the accuracy and consistency of the data largely depend on the diligence and expertise of the user, rather than on robust automated processes. This reliance on manual effort increases the likelihood of errors, duplication, and inconsistencies, which can propagate across different departments and processes.

Structurally, the existing system lacks a well-defined hierarchy for managing financial and organizational classifications, including account groups, cost centers, asset categories, and unit structures. This absence of structured relationships makes it challenging to maintain consistency across departments and results in discrepancies in reporting and operational data. For example, inconsistencies in account codes can lead to mismatched financial reports, while unstandardized asset categories may confuse asset tracking or depreciation calculations.

From a compliance perspective, the system provides inadequate support for audit trails and historical tracking of changes. This deficiency makes verification, reporting, and regulatory compliance a time-consuming and error-prone process. In addition, the system's usability is a major concern. The interface is outdated, non-responsive, and unintuitive, which hampers efficiency and increases reliance on extensive user training. New employees often struggle to navigate the system effectively, resulting in operational delays and reduced productivity.

Security is another critical limitation. Weak access controls and the absence of role-based permissions expose sensitive master data to potential unauthorized modifications. Furthermore, the system is not designed to scale with growing business requirements, limiting its ability to accommodate increased data volumes or future integration with advanced ERP modules.

Overall, the limitations of the existing ERP master data management system—ranging from poor automation, inconsistent data, and lack of standardization to weak security and limited scalability—significantly affect data integrity, operational efficiency, and compliance readiness. These challenges underscore the pressing need for a modern, centralized, and structured ERP master management solution, which can provide reliable, consistent, and secure master data to support all critical business processes.

2.3 Proposed System

To address the limitations of existing ERP master data management systems, the proposed system is a modern, web-based ERP Master Module that provides a centralized, structured, and secure platform for managing all critical master data. Unlike legacy systems, this solution emphasizes data accuracy, consistency, scalability, and user-friendliness. The system is developed using C# (.NET) for backend processing, Microsoft SQL Server for robust database management, and React-based frontend technologies for a responsive and intuitive user interface. This combination of technologies ensures efficient handling of large datasets while providing a smooth user experience.

The proposed system enforces strict validation rules and relational integrity to prevent duplicate or inconsistent records. For example, when adding a new account, the system automatically checks for existing entries and validates all required fields to maintain uniformity. A modular architecture allows each master module—such as organizational units, account codes, asset categories, and user management—to be maintained independently, facilitating future expansion and easier maintenance. Role-based access control ensures that only authorized personnel can modify sensitive data, enhancing security. Furthermore, cloud-based support via Microsoft Azure enables version control, centralized storage, and easy scalability for growing organizational needs.

Key advantages of the system include a centralized and normalized database, improved operational efficiency, accurate reporting, and compliance readiness. The modern, responsive interface reduces training time and enhances user productivity.

Advantages of the proposed system include:

- Centralized and normalized database
- Improved data accuracy and consistency
- Modern, responsive, and user-friendly interface
- Scalable and maintainable architecture
- Enhanced security and access control
- Improved operational efficiency and reporting reliability

2.4 System Requirements

The system has specific hardware and software requirements to ensure optimal performance.

2.4.1 Hardware Requirements

Processor	Intel i3 or higher
RAM	Minimum 8 GB
Storage	256 GB SSD or higher
Display Resolution	1366 × 768 or above

2.4.2 Software Requirements

Operating System	Windows 10 / Windows 11
Backend	C# (.NET)
Frontend	React, HTML5, CSS3, JavaScript
UI Frameworks	Bootstrap, Reactstrap
Database	Microsoft SQL Server
IDE	Visual Studio, Visual Studio Code
Repository & Cloud Platform	Microsoft Azure
Browser	Google Chrome / Microsoft Edge

3. System Design

3.1 Architectural Design

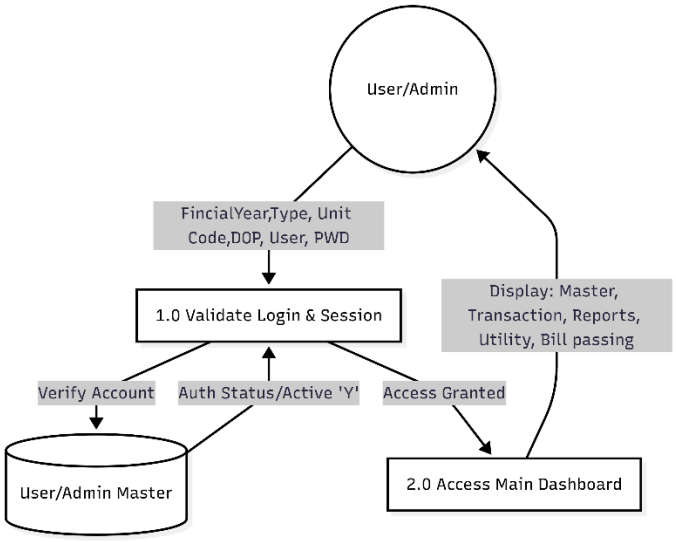
The system follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer. This architecture ensures separation of concerns, improved maintainability, and enhanced security.

3.1.1 Data Flow Diagram

The following DFD represents the flow of data in the system, showing Login, Main Page, Master hierarchy, Bank management, and Unit master interactions.

1. Login DFD Content (The Gateway)

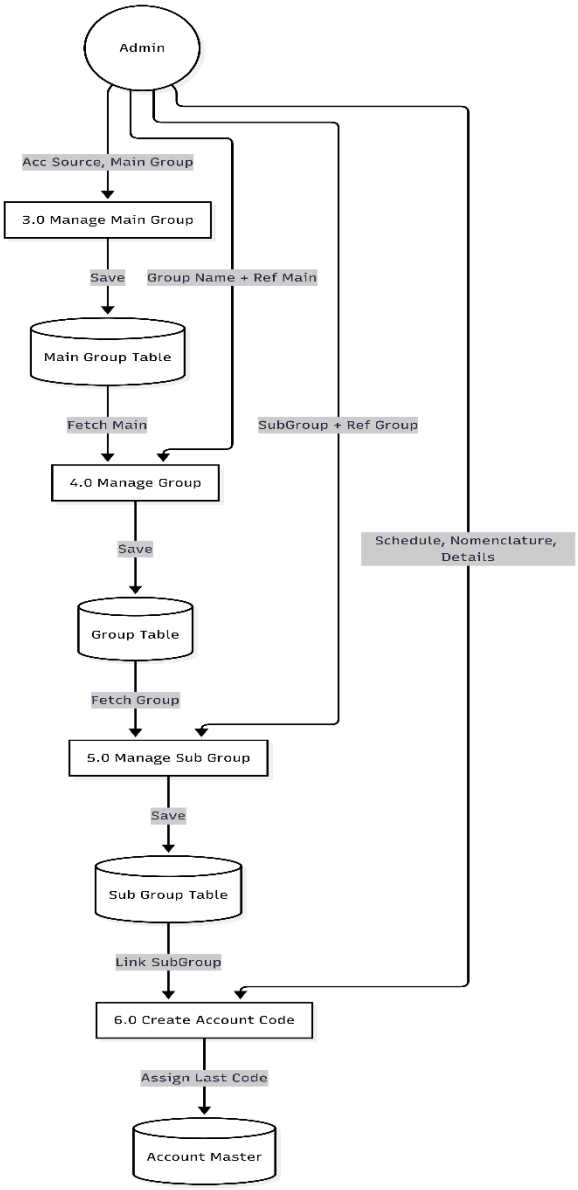
This module captures the initial session parameters. The flow ensures that the user is not just authenticated by password, but is also mapped to the correct financial context.



- Inputs: Financial Year, Type, Unit Code, Date of Processing (DOP), Username, and Password.
- Process: The system validates the User/PSD against the Staff Master. It then checks if the Unit Code is active and if the DOP falls within the selected Financial Year.
- Output: A valid session token containing the Entity ID and Unit Code, which acts as a global filter for all subsequent master pages.

2. Main Page & Master Hierarchy DFD Content

This represents the core organizational structure of your accounting system. The data flow follows a strict parent-child dependency.



- Main Group: The top-level category (e.g., Assets, Liabilities). It captures the Acc Source and Main Group Name.
- Group: Inherits the Main Group ID and adds a specific Group Name.
- Sub Group: Inherits both Main Group and Group IDs. This is where specific asset categories or expense types are defined (including depreciation rates like SLM/WDV found in your CSV files).

- Account Code Master: The final destination. It pulls the entire hierarchy (Source → Main → Group → Sub) and assigns specific Account Details, Schedules, and Nomenclature.
- Logic: The Last Code attribute ensures that the account numbering remains sequential within that specific sub-group.

3. Bank & Unit Master DFD Content

A. Bank Master (The Financial Data Repository)

This level captures the static details of the banking institution.

- Extended Data Points: Beyond account numbers, it captures IFSC/Swift Codes, Bank Branch specific contacts, and Overdraft (OD) Limits.
- Data Flow: It receives an Account Code from the Account Code Master. This creates a "Pointer" in the General Ledger so that any transaction involving this bank is automatically categorized under the correct accounting head.
- Validation: It ensures that no two banks share the same Bank Code within a single EntityID.

B. Bank Control (The Operational Logic Layer)

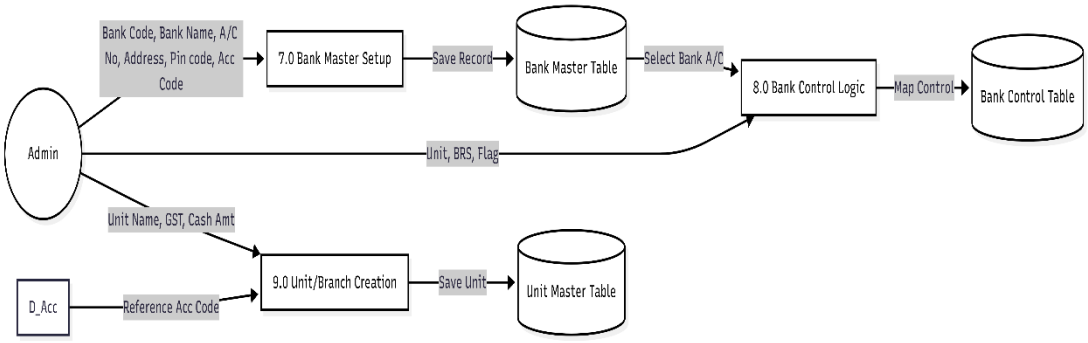
This acts as the "Traffic Controller" for financial operations.

- Functional Scope: It manages the User-Bank Mapping. It defines which users have the authority to view or process payments for specific accounts.
- BRS Logic: It tracks the "Last Reconciled Date." When a transaction occurs, the Bank Control layer checks the Active BRS flag to determine if the transaction requires immediate reconciliation or if it can be batched.
- Flags & Utype: The Utype (Usage Type) flag determines if the bank is used for Payroll, Vendor Payments, or Petty Cash, ensuring funds are not misallocated.

C. Unit Master (The Organizational Anchor)

The Unit Master is the root for all departmental data.

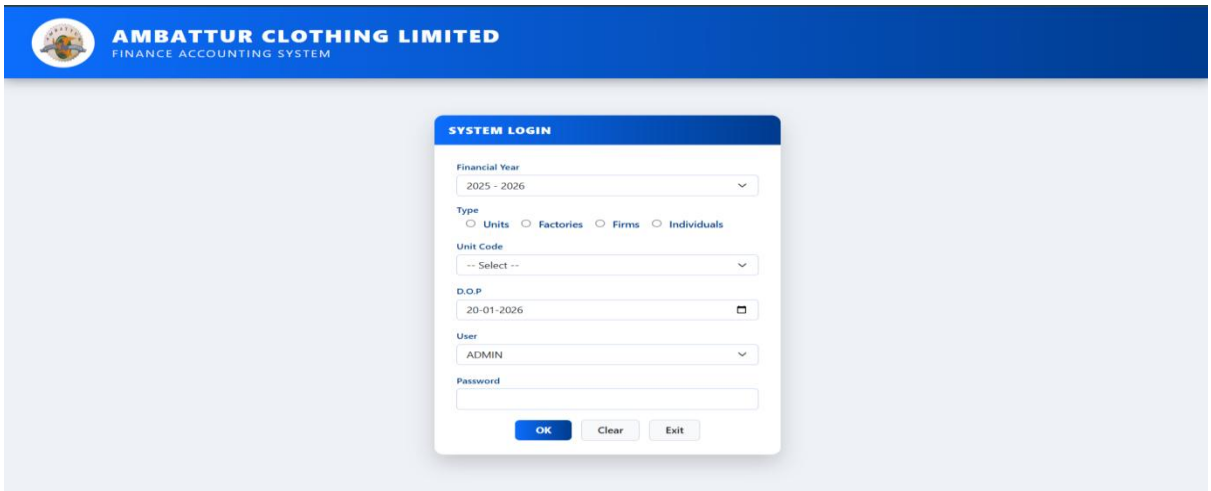
- Authorized Signatories: This module stores the digital identities or reference names of personnel authorized to approve vouchers. In the DFD, this flows into the Bill Passing and Transaction modules.
- Cash Management: It sets the Cash-in-Hand ceiling. If a unit attempts to record a cash transaction exceeding the Cash Amt limit defined here, the system triggers a "Control Violation" alert.
- GST & Compliance: It acts as the central point for Tax Reporting. Since GST is often unit-specific (based on location), this master ensures that the correct tax state-code is applied to every invoice generated by that specific branch.

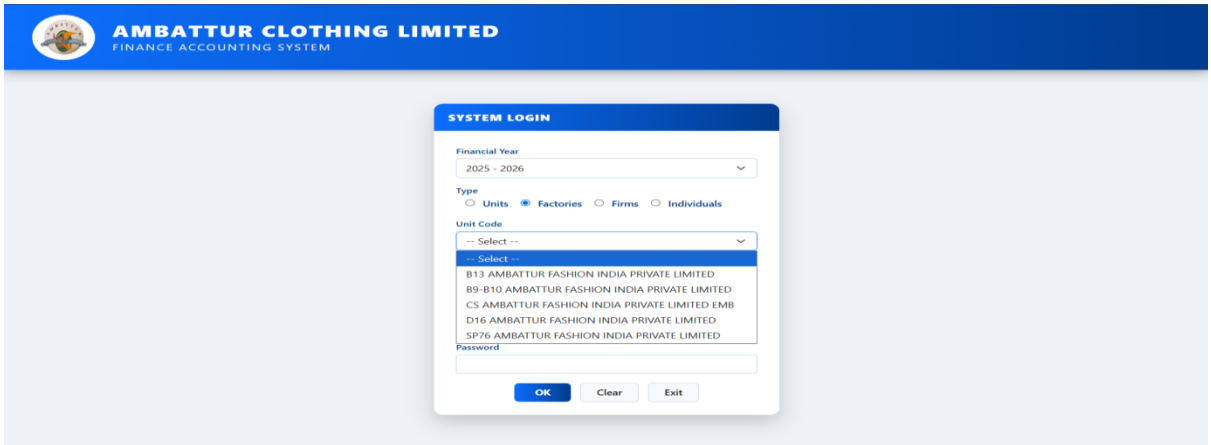


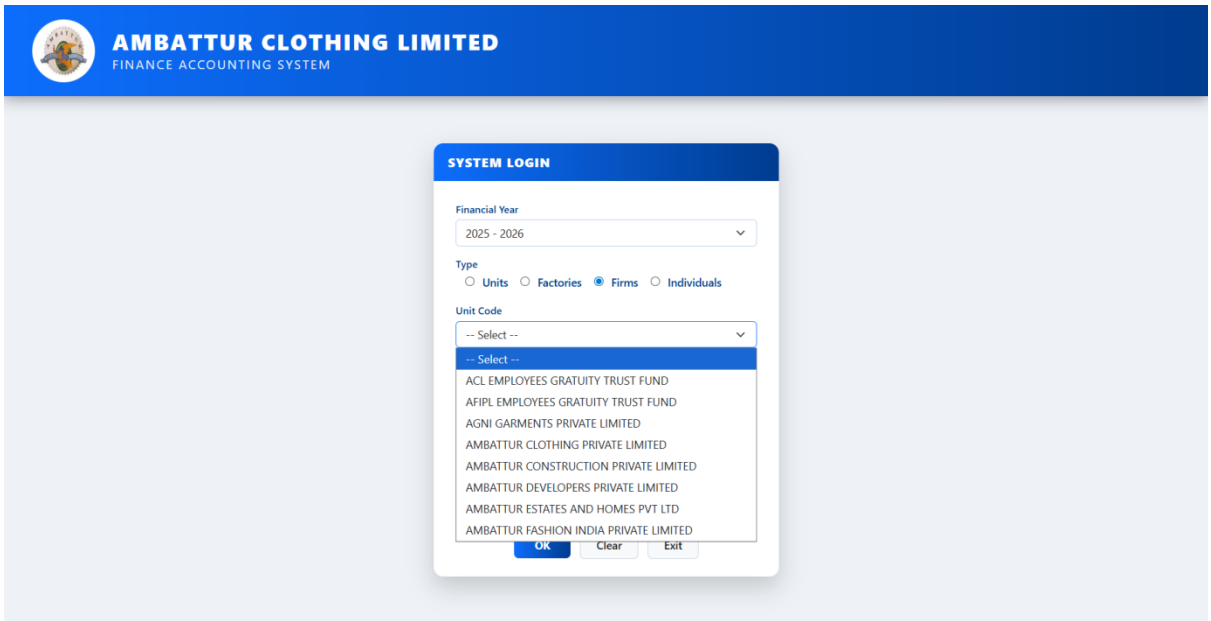
3.2 GUI Design

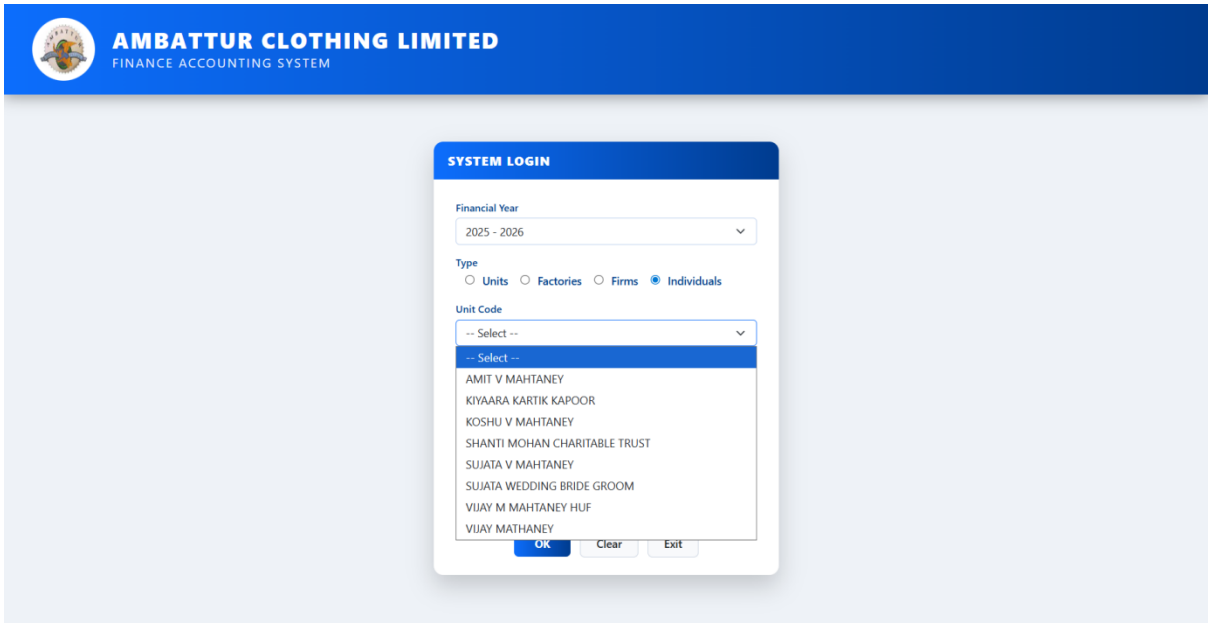
The graphical user interface is designed using React and Reactstrap, ensuring responsiveness and consistency. The interface follows ERP standards with structured forms, dropdown controls, and tabular data presentation.

LOGIN:

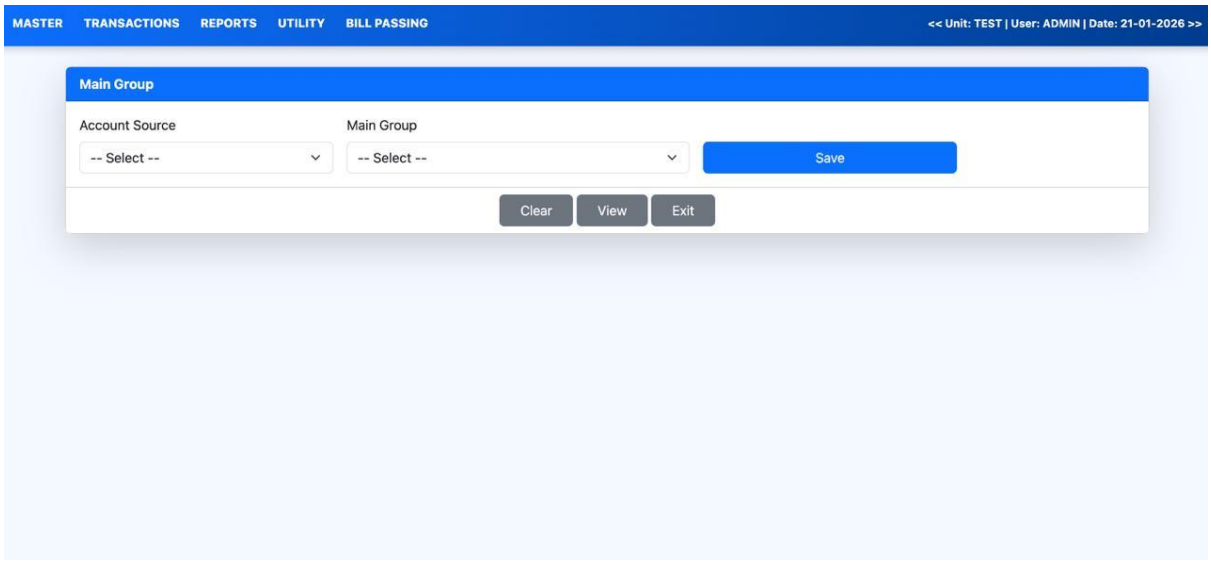


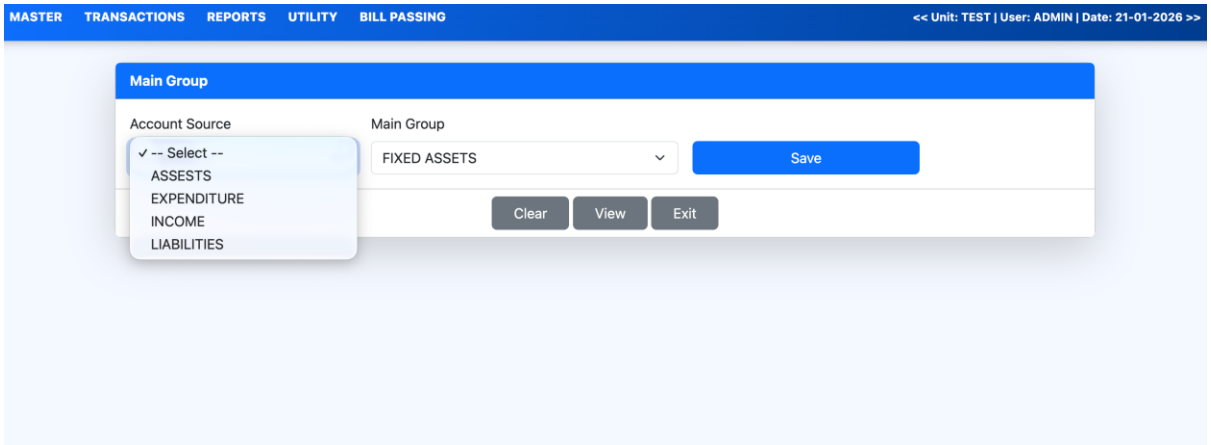

AMBATTUR CLOTHING LIMITED
FINANCE ACCOUNTING SYSTEM
SYSTEM LOGIN
Financial Year
2025 - 2026
Type
Units
Factories
Firms
Individuals
Unit Code
-- Select --
-- Select --
B13 AMBATTUR FASHION INDIA PRIVATE LIMITED
B9-B10 AMBATTUR FASHION INDIA PRIVATE LIMITED
CS AMBATTUR FASHION INDIA PRIVATE LIMITED EMB
D16 AMBATTUR FASHION INDIA PRIVATE LIMITED
SP76 AMBATTUR FASHION INDIA PRIVATE LIMITED
Password
OK
Clear
Exit

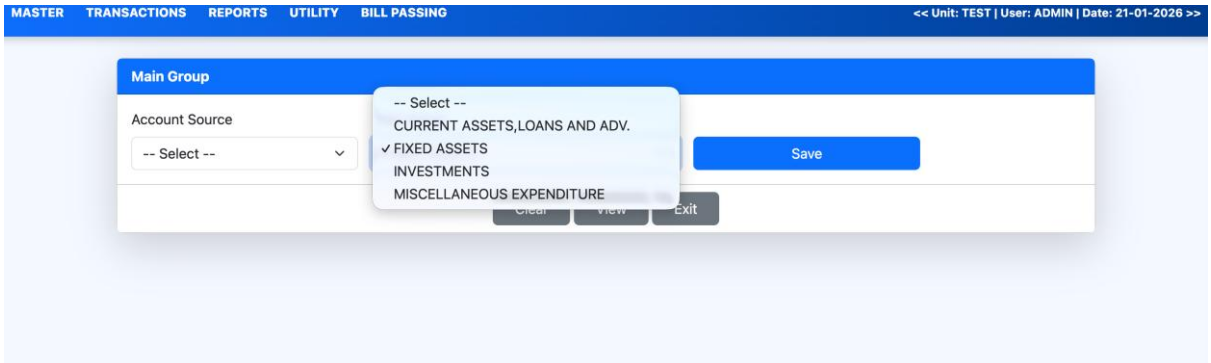

AMBATTUR CLOTHING LIMITED
FINANCE ACCOUNTING SYSTEM
SYSTEM LOGIN
Financial Year
2025 - 2026
Type
Units
Factories
Firms
Individuals
Unit Code
-- Select --
-- Select --
ACL EMPLOYEES GRATUITY TRUST FUND
AFIPL EMPLOYEES GRATUITY TRUST FUND
AGNI GARMENTS PRIVATE LIMITED
AMBATTUR CLOTHING PRIVATE LIMITED
AMBATTUR CONSTRUCTION PRIVATE LIMITED
AMBATTUR DEVELOPERS PRIVATE LIMITED
AMBATTUR ESTATES AND HOMES PVT LTD
AMBATTUR FASHION INDIA PRIVATE LIMITED
OK
Clear
Exit

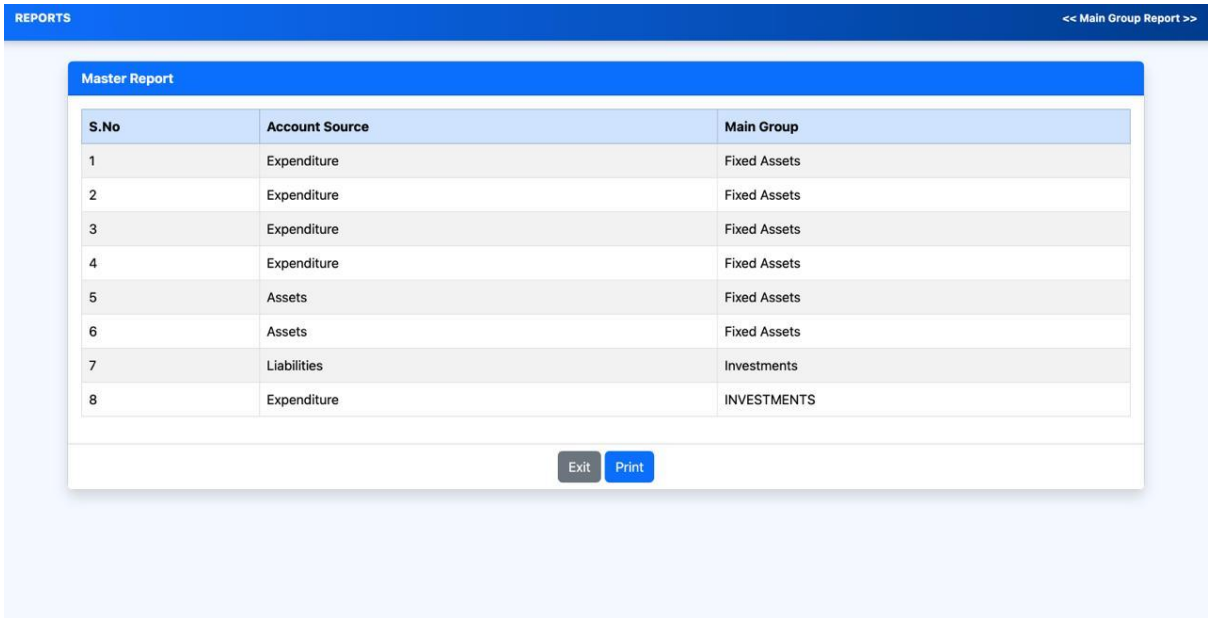

AMBATTUR CLOTHING LIMITED
FINANCE ACCOUNTING SYSTEM
SYSTEM LOGIN
Financial Year
2025 - 2026
Type
Units
Factories
Firms
Individuals
Unit Code
-- Select --
-- Select --
AMIT V MAHTANEY
KIYAARA KARTIK KAPOOR
KOSHU V MAHTANEY
SHANTI MOHAN CHARITABLE TRUST
SUJATA V MAHTANEY
SUJATA WEDDING BRIDE GROOM
VIJAY M MAHTANEY HUF
VIJAY MATHANEY
OK
Clear
Exit

MAIN GROUP MASTER:



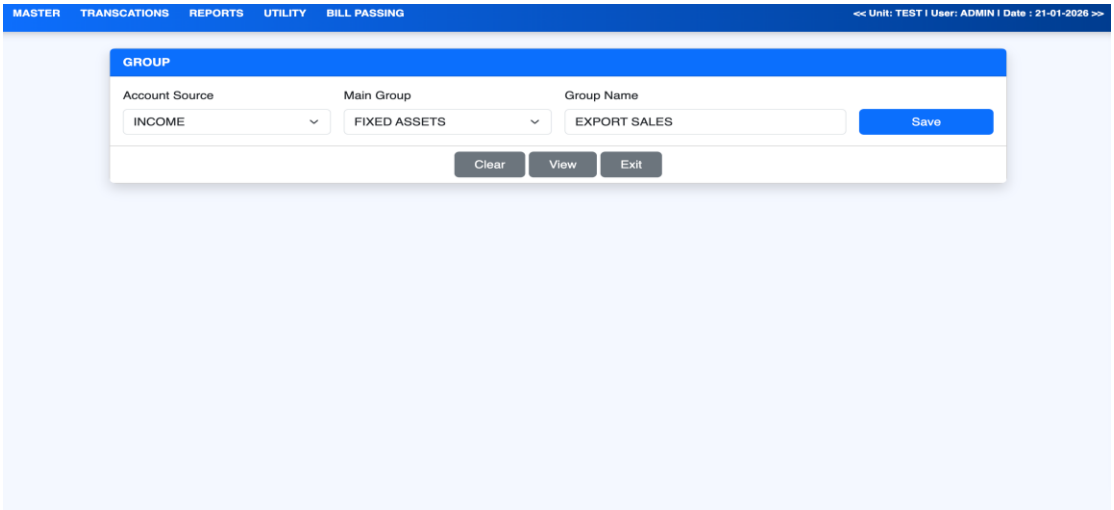


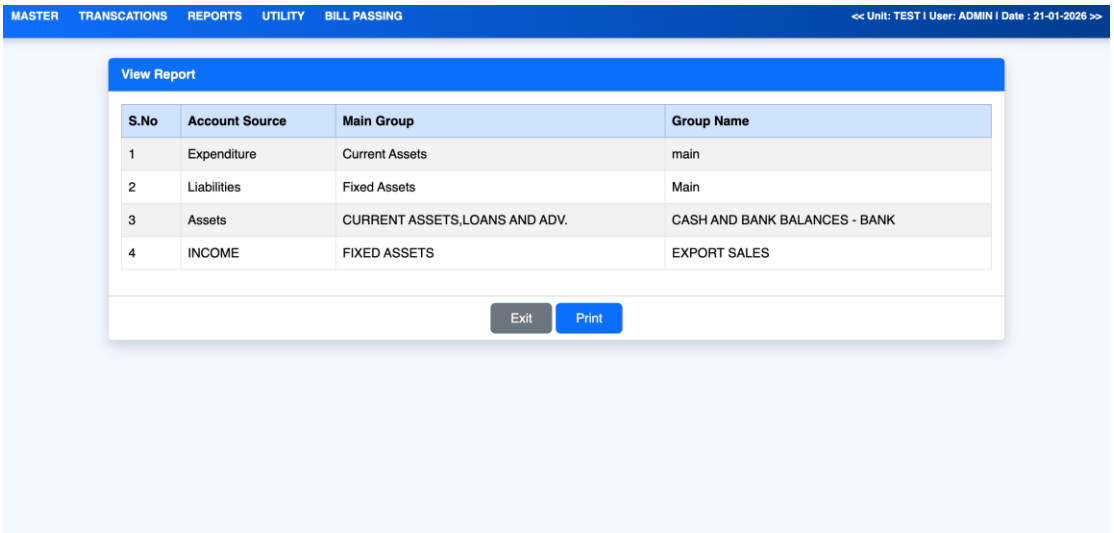




S.No	Account Source	Main Group
1	Expenditure	Fixed Assets
2	Expenditure	Fixed Assets
3	Expenditure	Fixed Assets
4	Expenditure	Fixed Assets
5	Assets	Fixed Assets
6	Assets	Fixed Assets
7	Liabilities	Investments
8	Expenditure	INVESTMENTS

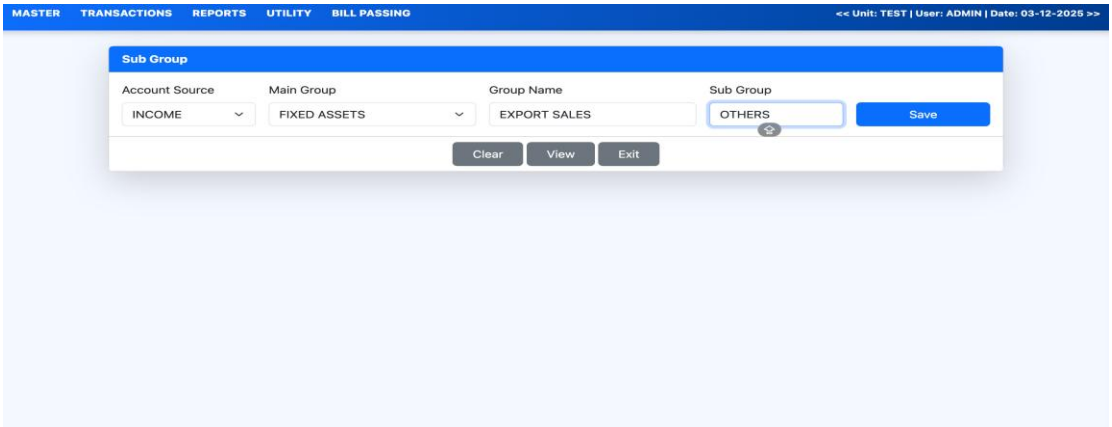
GROUP MASTER:

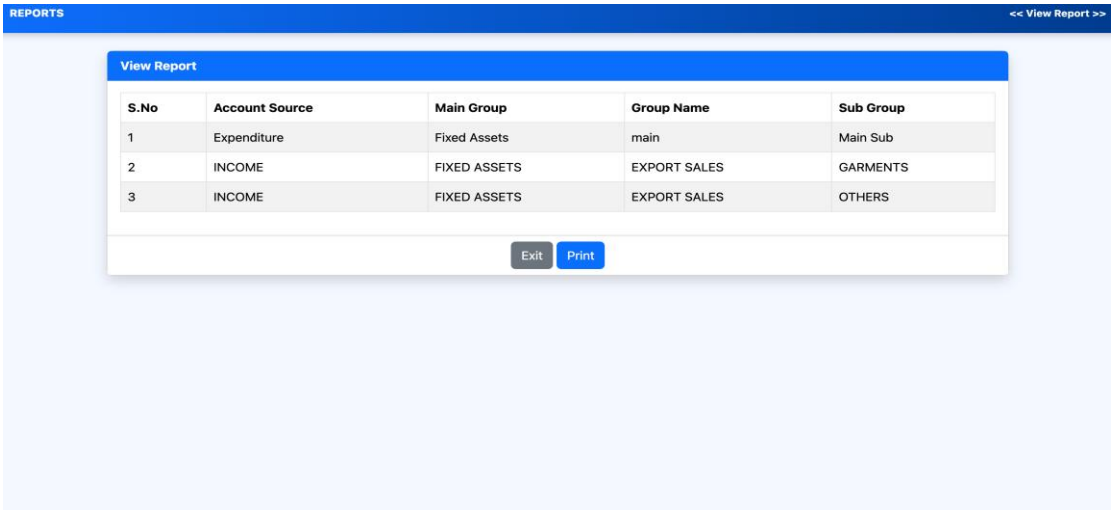




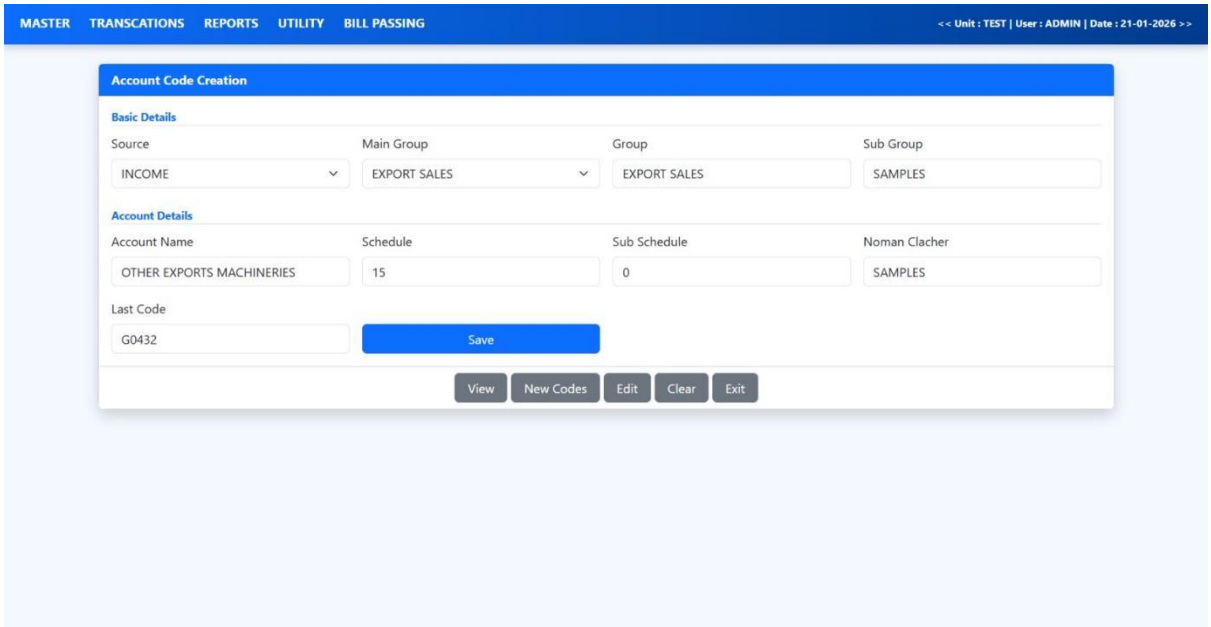
S.No	Account Source	Main Group	Group Name
1	Expenditure	Current Assets	main
2	Liabilities	Fixed Assets	Main
3	Assets	CURRENT ASSETS,LOANS AND ADV.	CASH AND BANK BALANCES - BANK
4	INCOME	FIXED ASSETS	EXPORT SALES

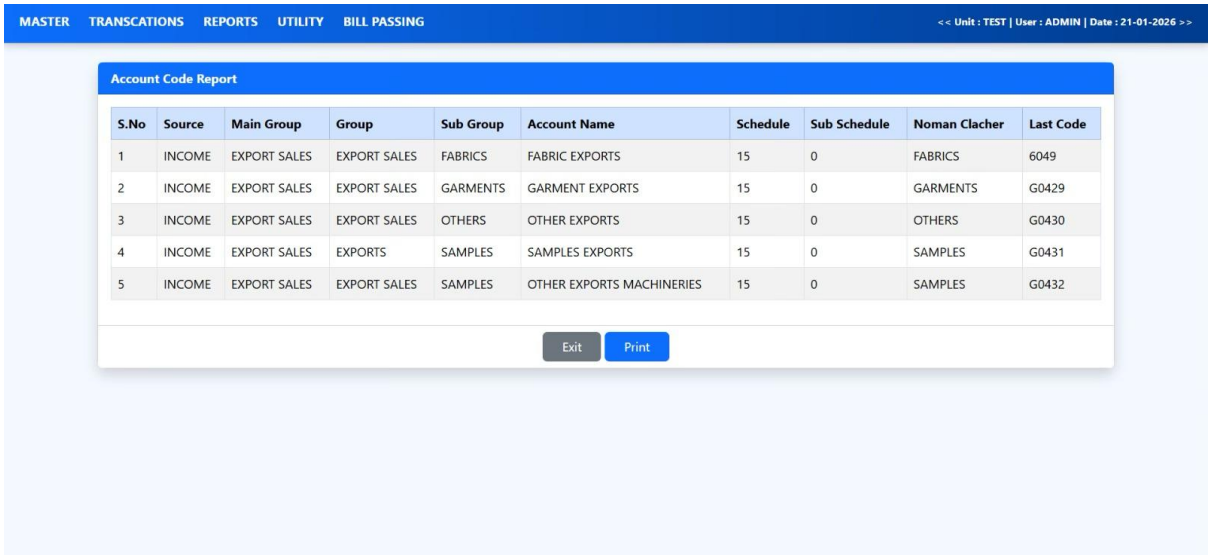
SUB GROUP MASTER:





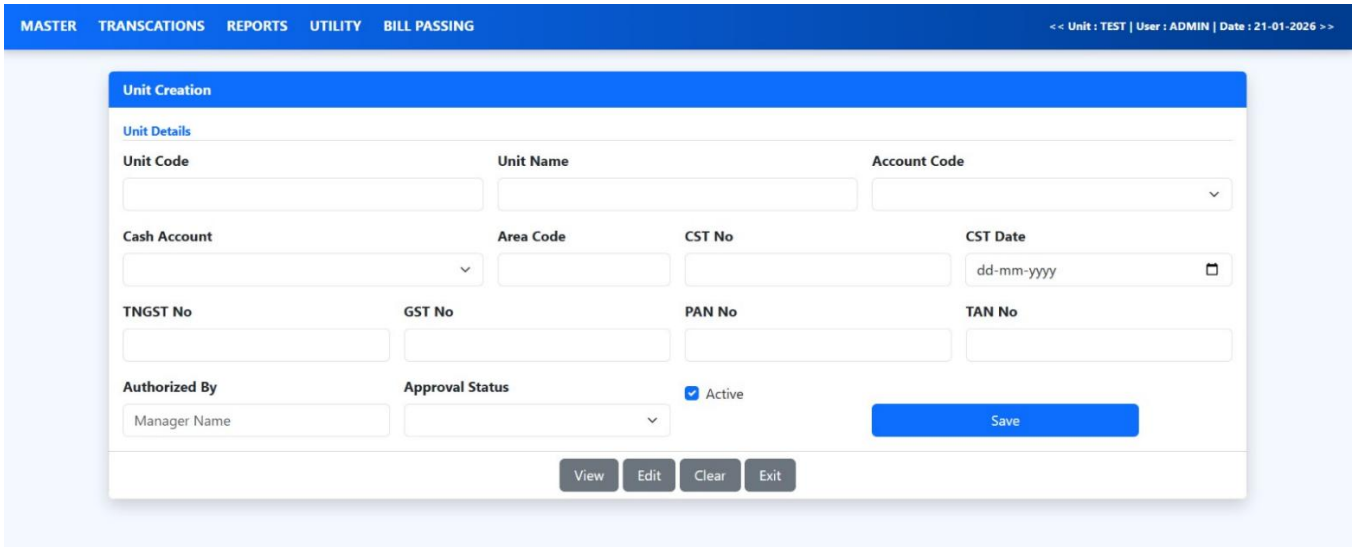
ACCOUNT CODE MASTER:

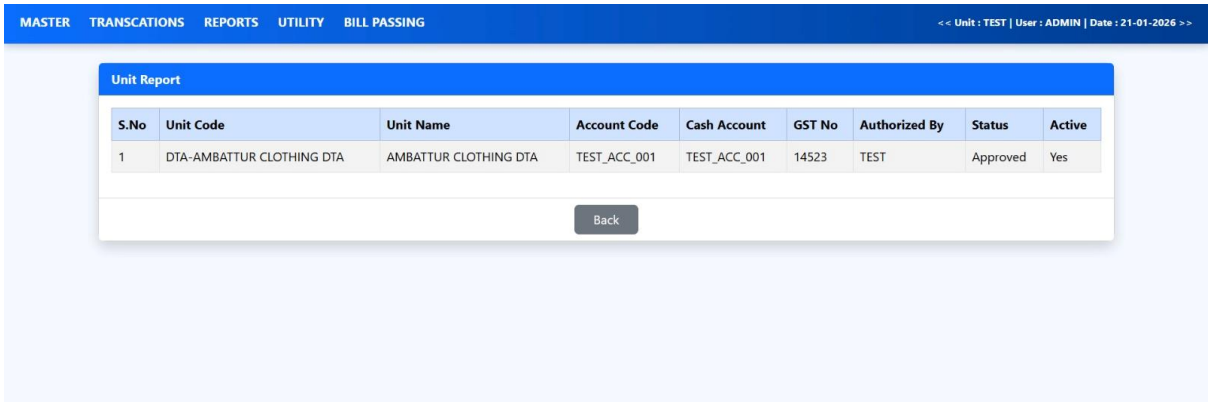




S.No	Source	Main Group	Group	Sub Group	Account Name	Schedule	Sub Schedule	Noman Clacher	Last Code
1	INCOME	EXPORT SALES	EXPORT SALES	FABRICS	FABRIC EXPORTS	15	0	FABRICS	6049
2	INCOME	EXPORT SALES	EXPORT SALES	GARMENTS	GARMENT EXPORTS	15	0	GARMENTS	G0429
3	INCOME	EXPORT SALES	EXPORT SALES	OTHERS	OTHER EXPORTS	15	0	OTHERS	G0430
4	INCOME	EXPORT SALES	EXPORTS	SAMPLES	SAMPLES EXPORTS	15	0	SAMPLES	G0431
5	INCOME	EXPORT SALES	EXPORT SALES	SAMPLES	OTHER EXPORTS MACHINERIES	15	0	SAMPLES	G0432

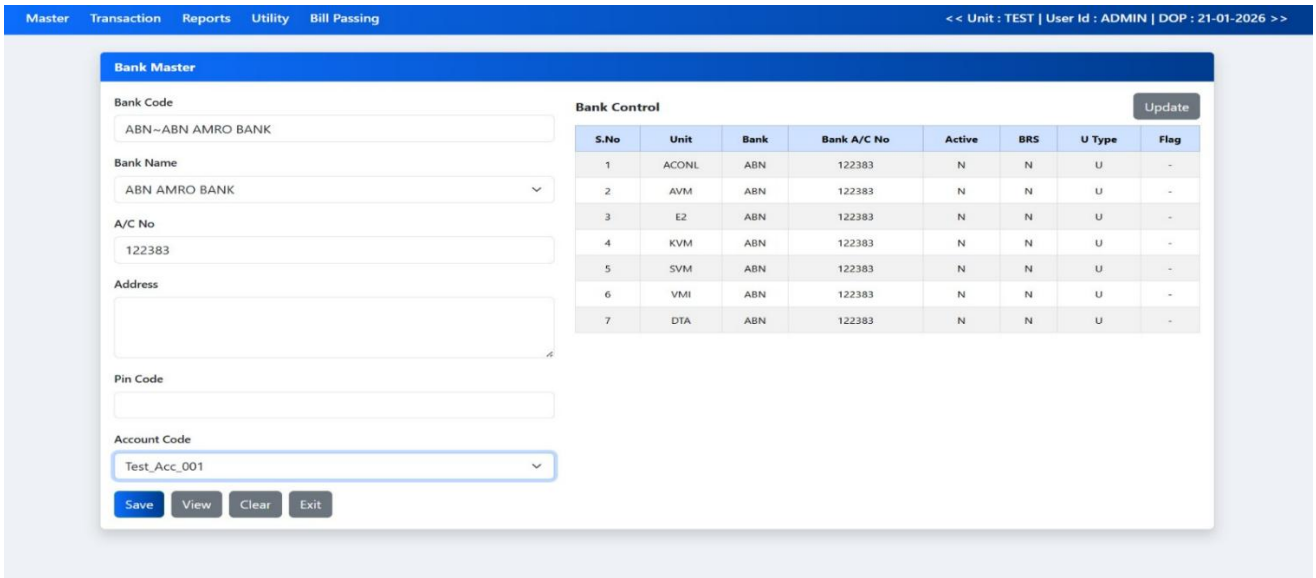
UNIT CREATION

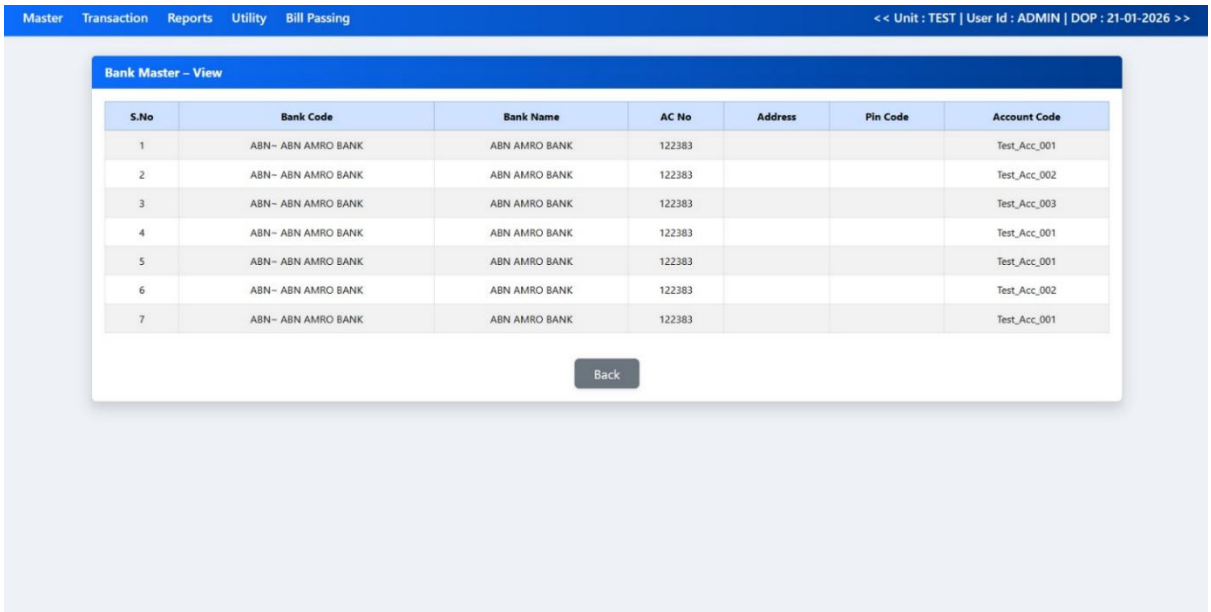




S.No	Unit Code	Unit Name	Account Code	Cash Account	GST No	Authorized By	Status	Active
1	DTA-AMBATTUR CLOTHING DTA	AMBATTUR CLOTHING DTA	TEST_ACC_001	TEST_ACC_001	14523	TEST	Approved	Yes

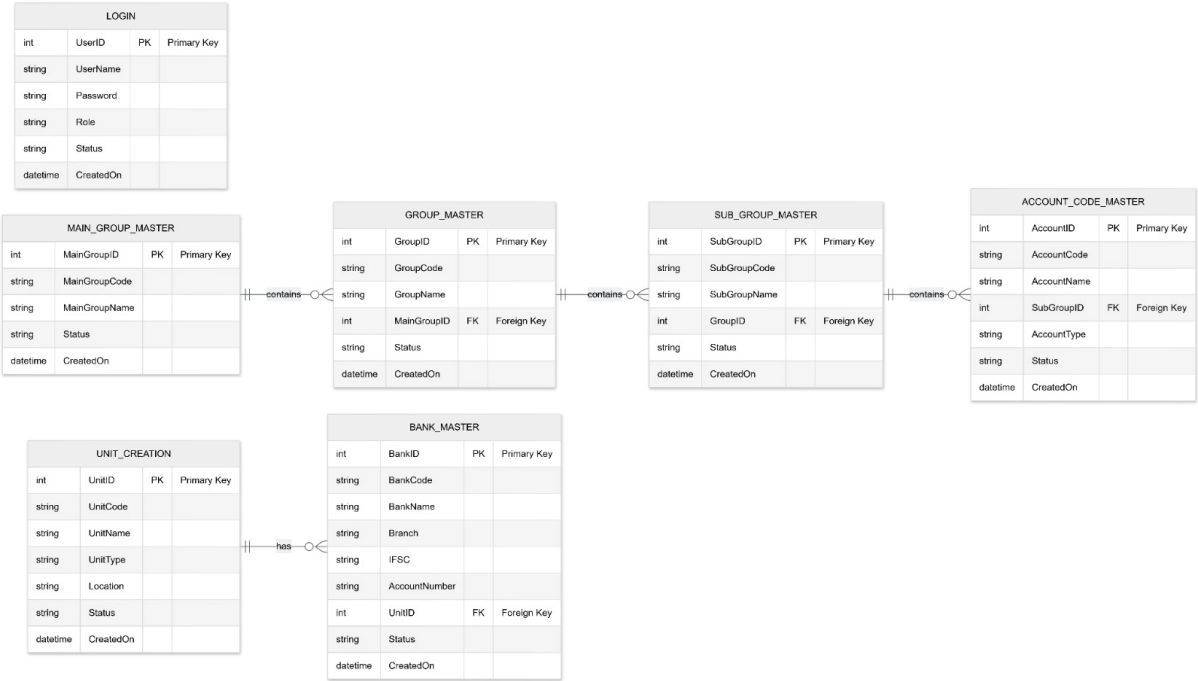
BANK MASTER:





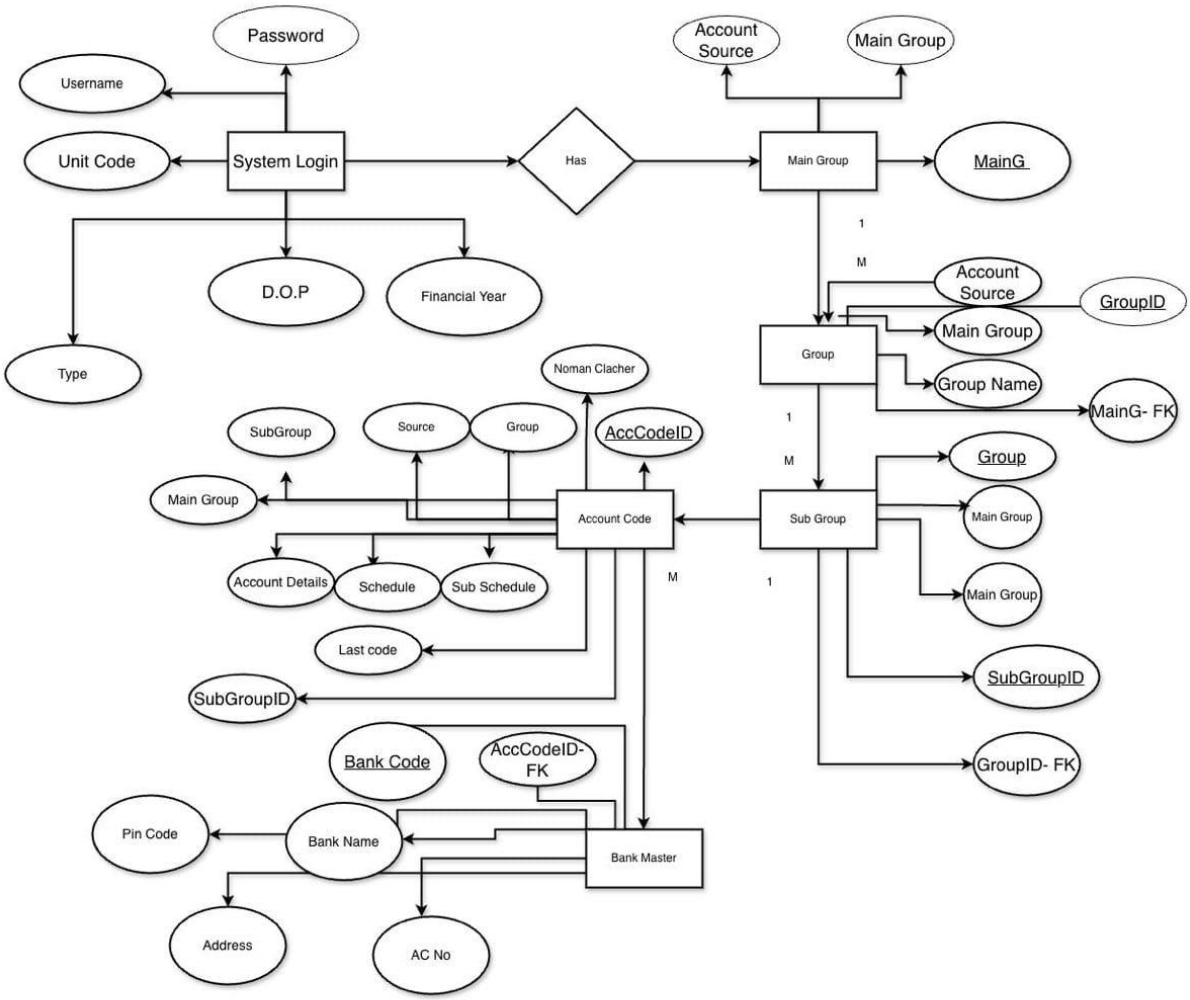
3.3 UML DIAGRAM

The UML (Unified Modeling Language) structure represents a high-integrity financial system where Login sessions are anchored by Financial Year and UnitCode. Data flows through a nested hierarchy—from Main Group to Account Master—ensuring strict classification. These accounts link to Bank and Unit Masters, governed by Bank Control logic for operational compliance.



Module	Primary Key / Index	Key Attributes for UML
Login	UserID	Financial Year, UnitCode, DOP
Main Group	MainCode	Acc Source, EntityID
Group	GroupCode	Parent MainCode, GroupName
Sub Group	SubCateCode	DepnType, DepnRate, Category
Account Master	AccCode	Schedule, Nomenclature, LastCode
Bank Master	BankCode	A/C No, IFSC, Address, Pin
Bank Control	ControlID	UnitCode, BRS Flag, Utype
Unit Master	UnitCode	GST No, Cash Limit, Status

3.4 ER DIAGRAM



The ER (Entity-Relationship) architecture establishes a structured data flow beginning with Login authentication, where a user's session is bound to a specific Financial Year and Unit. This initiates a strictly hierarchical master data tree: Main Groups define primary accounting sources, which cascade into Groups, then granular Sub-Groups (containing depreciation logic), and finally unique Account Codes.

These Account Codes serve as the bridge to operational entities, linking directly to the Bank Master and Unit Master. Financial integrity is maintained via Bank Control logic, which manages unit-specific bank reconciliations and usage flags, ensuring every transaction aligns with systemic compliance and organizational limits.

4. Project Description

The ERP Master Module is designed as a centralized and comprehensive solution for managing core master data essential for enterprise operations. Master data acts as the foundation for all ERP functionalities, including finance, asset management, human resources, procurement, and reporting. By providing a unified platform to create, maintain, and validate critical data, the ERP Master Module ensures consistency, accuracy, and integrity across all organizational processes.

The system manages a wide range of master data components, including organizational units, account codes, asset classifications, supplier and customer information, banking details, and user credentials. Each component is structured hierarchically and validated against predefined rules, ensuring that data entered into the system adheres to organizational standards and avoids redundancy or duplication. This structured approach enables the organization to maintain a single source of truth, which is critical for operational efficiency and informed decision-making.

The ERP Master Module also establishes the groundwork for integration with future ERP modules. For example, finance and accounting modules rely heavily on accurate account codes and financial groupings, while asset management modules depend on precise asset classifications and tracking information. Reporting and business intelligence functions are also supported by the module, as clean and consistent master data enables accurate analytics, compliance reporting, and strategic planning.

In addition, the module incorporates role-based access control, ensuring that only authorized personnel can modify sensitive master data. This enhances security and governance while providing audit trails for compliance purposes. The modern, web-based interface, built using React and Bootstrap, ensures responsiveness, usability, and minimal training requirements, allowing employees to efficiently perform their tasks.

Overall, the ERP Master Module acts as the backbone of the organization's ERP ecosystem. By centralizing, standardizing, and securing master data, it significantly improves operational efficiency, reduces errors, supports regulatory compliance, and provides a scalable foundation for future ERP development and business growth.

5. System Development

The development of the ERP Master Module involved a combination of modern software technologies, database systems, and frontend frameworks to ensure a secure, scalable, and user-friendly solution. The system is designed as a web-based application, providing centralized access to master data while maintaining data integrity and operational efficiency across the enterprise.

5.1 Languages and Tools Used

The backend of the system is developed using C# (.NET), which offers a robust environment for creating scalable, object-oriented applications with strong type-checking and support for multi-layered architecture. Microsoft SQL Server serves as the relational database management system, providing a secure, normalized, and efficient structure for storing master data. SQL Server ensures data integrity through relational constraints, indexing, and transaction management.

For the frontend, React.js is utilized to build a responsive, modular, and dynamic interface. Combined with HTML5, CSS3, and JavaScript, the interface provides interactive forms, structured tables, and intuitive navigation. Bootstrap and Reactstrap frameworks enhance the UI design by offering pre-built, responsive components that adhere to modern ERP interface standards, reducing development time and improving user experience.

The system also integrates Microsoft Azure for cloud-based repository support, enabling version control, centralized storage, scalability, and secure remote access. Azure ensures that master data can be maintained and accessed reliably across multiple locations while supporting future expansion and integration with additional ERP modules.

5.2 Development Workflow and Pseudocode

The development workflow follows a structured approach. The system first initializes the environment and authenticates users through role-based login. Upon successful authentication, the dashboard displays all available master modules. Users can select a module, input relevant master data, and submit it. Each entry undergoes validation against predefined rules to prevent duplication, enforce standardization, and maintain relational integrity. Once validated, the data is securely stored in the database. If authentication fails or validation errors occur, the system displays appropriate error messages to guide the user.

The pseudocode summarizes this process:

1) **START**
2) **Initialize system**
3) **Authenticate user**
4) **IF authentication successful THEN**
5) **Display dashboard**
6) **FOR each master module**
7) **Accept input**
8) **Validate data**
9) **Store in database**
10) **END FOR**
11) **ELSE**
12) **Display error message**
13) **END IF**
14) **STOP**

This structured approach ensures that the ERP Master Module provides a reliable, secure, and efficient solution for managing enterprise master data, laying a strong foundation for future ERP development and operational excellence.